MIL-I-48252A (AR)
1 December 1977
SUPERSEDING
MIL-I-48252 (MU)
14 September 1973

MILITARY SPECIFICATION

IGNITER TUBES

This specification is approved for use by the U.S. Army Armament Research and Development Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope.-This specification covers igniter tubes for use with 155MM and 8 Inch Propelling Charges.

2. APPLICABLE DOCUMENTS

2.1 Issues of Documents.- The following documents of the issue in effect on date of invitation for bids or request for proposal, forms a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification for

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-286 - Propellants, Solid, Sampling, Examination and Testing

FSC: 1320

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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DRAWINGS (See 6.5)

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND

8799112 - Marking Diagram and Sealing for Inter-Plant or Conus Shipment of Ammunition Fiberboard Boxes
9247366 - Composition
9282963 - Packing & Marking for Box, Fiberboard, for Igniter, Tube for 155MM & 8-Inch Propelling Charges
9289172 - Tube
9285215 - Tube
9285248 - Tube
9325882 - Tube

Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer.

3. REQUIREMENTS

3.1 Materials.-The constituent materials shall comply with the requirements of all applicable drawings and specifications (See 6.4).

3.2 Parts.-Igniter tubes shall comply with all requirements specified in Drawing (dwg.) 9247366, all applicable specifications and the applicable drawings specified below.

3.2.1 Dwg. 9285215 for 155MM Propelling Charge, M203A1.

3.2.2 Dwg. 9285248 for 8-Inch Propelling Charge, M188A1.

3.2.3 Dwg. 9325882 for 155MM Propelling Charge, M119A1.

3.2.4 Dwg. 9289172 for 155MM Propelling Charge, 155MM PXR-6255.

3.3 Stability.-The nitrocellulose in the igniter tube shall not completely change the color of the methyl violet paper to salmon pink color in less than thirty-five (35) minutes when tested according to 4.4.3.1.

3.4 Weight.-The weight of the igniter tube shall be as specified below when tested in accordance with 4.4.2.2.

3.4.1 Dwg. 9285215. The weight shall be 140 $\pm$ 7 grams.

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3.4.2 Dwg. 9285248. The weight shall be 147 ± 7 grams.

3.4.3 Dwg. 9325882. The weight shall be 124 ± 6 grams.

3.4.4 Dwg. 9289172. The weight shall be 136 ± 7 grams.

3.5 Cracks and Soft Spots.-All parts shall be free of cracks and soft spots.

3.6 Density.-The density shall not be less than 0.53 grams per cubic centimeter.

3.7 Workmanship.-All parts shall be fabricated in a thorough, workmanlike manner. Each tube shall be free of molding flaws, dents, warpage outside the material envelope, folds, bends, tears, flat areas, frayed edges, foreign material greater than 1/4 inch in any dimension and delaminations.

3.8 First Article Inspection.-This specification makes provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection and Standard Quality Assurance Provisions.-Unless otherwise specified herein or in the Contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of Inspections.-The following types of inspection shall be conducted on this item:

a. First Article Inspection
b. Quality Conformance Inspection

4.3 First Article Inspection

4.3.1 Submission.-The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.3.2. The first article sample shall consist of 25 tubes.

4.3.2 Inspection to be Performed.-The First Article parts submitted in accordance with paragraph 4.3.1 will be subjected by the Government to any or all the requirements of the applicable drawings and to the inspection and test of this specification.

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4.3.3 Rejection.-See MIL-A-48078.

4.4 Quality Conformance Inspection

4.4.1 Inspection Lot Formation.-Inspection lots shall comply with the lot formation provisions of MIL-A-48078. In addition, inspection lots igniter shall contain:

a. Nitrocellulose from one manufacturer and from not more than one production lot.

b. Diphenylamine from one manufacturer and from not more than one production lot.

c. Acrylic fibers from one manufacturer and from not more than one production lot.

d. Resin from one manufacturer.

e. Kraft from one manufacturer and not more than one production lot.

4.4.2 Examination.-See MIL-A-48078.

a. Sampling Plans. Unless otherwise specified in the Classification of Defects and Test Tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.1	Nitrocellulose		NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical 1	Stability			3.3	4.4.3.1
Major	None defined				
Minor	None defined				

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

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PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER (See 3.2) / NEXT HIGHER ASSEMBLY
4.4.2.2	Tube, Igniter				
CATEGORY	**EXAMINATION OR TEST**	**NO. OF SAMPLE UNITS**	**AQL OR 100%**	**REQUIREMENT PARAGRAPH**	**PARAGRAPH REFERENCE /INSPECTION METHOD**
Critical	None defined				
Major					
101	Cracks and soft spots		100%	3.5	Visual
102	Total length		100%	3.2	Gage
103	Straightness		100%	3.2	Gage
104	Location of holes		0.40%	3.2	Gage
105	Weight		0.40%	3.4	4.5.2
106	Density	10		3.6	4.4.3.1
107	Composition	10		3.2	4.4.3.2
108	Wall thickness		0.40%	3.2	Gage
109	Outside Diameter		0.40%	3.2	Gage
Minor					
201	Workmanship		0.65%	3.7	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-I-48252A (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.3	Sealed Barrier Bag	1 OF 1	9282963
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Bag punctured, torn or cut		0.40%	3.2	Visual
102	Seal mislocated or incomplete		0.40%	3.2	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-I-48252A (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.4	Box, Sealed				9282963
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major	None defined				
Minor					
201	Sealing strip torn, incomplete, badly wrinkled or otherwise fails to seal box		0.65%	3.2	Visual
202	Box torn, cut or punctured		0.65%	3.2	Visual
203	Marking missing, misleading or unidentifiable		0.65%	3.2	Visual

NOTES:

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4.4.3 Testing

4.4.3.1 Stability (See 3.3).-A stability heat test at 135.5°C shall be performed in accordance with MIL-STD-286, Method 404.1.2. A random sample of 10 tubes shall be selected from the first article sample and the first three production lots to provide the required test specimens. Failure of any specimen is cause for lot rejection. When three consecutive lots have been accepted, sampling may be reduced to 5 tubes from each lot. However, if a failure occurs, sampling shall revert to 10 per lot until 3 consecutive lots have been accepted.

4.4.3.2 Density.-One inch wide ring specimens shall be randomly selected near each end and every seven (7) inches along the length of the tube. The tube shall be from the weight sample and retained in the container. The wall thickness shall fall within the tolerance of the drawing. Each specimen shall be weighed to the nearest 0.01 gram and the dimensions determined to the nearest 0.001 inch. The densities shall be calculated from these values. If any sample fails to comply with the requirement the lot shall be rejected.

4.4.3.3 Composition (See dwg. 9247366).-Ten (10) tubes shall be randomly selected from the first article sample and from each of the first three production lots to provide the required test specimens. Failure of any specimen is cause for lot rejection. When three consecutive lots have been accepted, sampling may be reduced to five from each lot. If a failure occurs, sampling shall revert to ten per lot until three consecutive lots have been accepted. Duplicate specimens shall be obtained from each sample for analysis in accordance with 4.5.1.

4.4.4 Inspection Equipment.-The inspection equipment required to perform the examinations and tests prescribed herein is described in the Paragraph Reference/Inspection Method Column in the tables starting with paragraph 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See section 6 of MIL-A-48078.

4.5 Test Methods and Procedures

4.5.1 Composition

4.5.1.1 Preparation of Specimen (See 4.3.3.3).-Each specimen shall be divided into 1/8 inch (approximate) cubes or equivalent. The specimens shall then be thoroughly mixed to form a composite. From the composite, approximately 1.0 g shall be selected as the analysis sample and weighed to the nearest 0.2 milligram (mg), (Weight A) in an extraction thimble. The same thimble should be carried through the extractions in 4.5.1.3.1 and 4.5.1.3.3.

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4.5.1.2 Total Volatiles (See 4.3.3.4).-The samples from 4.5.1.1 shall be dried for 3 hours at 100 ± 2°C, cooled in a desiccator (1/2 hour min.), and weighed to the nearest 0.2 mg (Weight W). Calculate the percent total volatiles as follows:

$$\% \text{ Total Volatiles} = \frac{A - W \times 100}{A}$$

4.5.1.3 Analytical Procedure

4.5.1.3.1 Extraction.-Using a tared, extra-coarse sintered glass thimble, extract each sample from 4.5.1.2, with methylene cloride, for a minimum of 16 hours in a Roweg Extractor (MIL-STD-286, Method 104.1). Evaporate the extract to dryness, using a current of clean, dry air. This extract is to be used for diphenylamine determination in 4.5.1.3.2. Dry the residues (non-extractable material) in the extraction thimbles by aspirating the thimble and contents until the odor of methylene chloride is not discernible. Dry the thimble plus residue for 1/2 hour minimum at 100 ± 2°C. Cool in a desiccator (1/2 hour min.). Determine the dry weight of this residue (Weight C) to the nearest 0.2 mg as the difference between this weight and the weight of the empty thimble. This residue is to be used for the nitrocellulose determination in 4.5.1.3.3.

4.5.1.3.2 Diphenylamine (DPA).-To the dried extract from 4.5.1.3.1, add 50 milliliters (ml) of glacial acetic acid. If necessary, warm slightly to dissolve the extract, then cool to room temperature. Complete the determination of diphenylamine in the solution according to Method 201.1 (starting with paragraph 5.5) of Standard MIL-STD-286, except that 25 ml of methylene chloride is to be added after addition of the potassium iodide solution. When the end point appears to have been reached, (absence of blue color), the flask is stoppered and shaken. If no color appears, the end point has been reached; otherwise, continue titration until the blue color is discharged and shaking produces no further color. Calculate the percent DPA as shown in Standard MIL-STD-286, Method 201.4, paragraph 5.11.

4.5.1.3.3 Nitrocellulose.-To provide good heat conduction to the solvent, place a sufficient quantity of glycerin to the flask bottom and the surface of the heat source to produce a continuous film. Add 10 ml of ethyl alcohol to the thimble and residue from 4.5.1.3.1 and extract with acetone using the method in 4.5.1.3.1. After extraction, dry the residue remaining at 100 ± 2°C for one hour. Cool in a desiccator (1/2 hour min.), weigh to the nearest 0.2 mg (Weight D) and save for the acrylic determination in 4.5.1.3.4. Evaporate the acetone extract to dryness and add 25 ml of n-butyl acetate and 45 ml of glacial acetic

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acid. Replace the air with a stream of inert gas (CO_2 or N_2). Maintain the inert atmosphere throughout this test. Reflux the solution for five (5) minutes. Add to the flask, via the condenser, 25 ml of 0.7N ferrous ammonium sulfate (prepared per MIL-STD-286, Method 604.1) and 10 ml concentrated hydrocloric acid. Reflux for thirty (30) minutes more to obtain a yellow liquid in which the specimen is completely disintegrated. With the inert gas still running, cool in a water bath to room temperature. Add 5 ml of 20% ammonium thiocyanate solution (MIL-STD-286, Method 601.1). Titrate with a 0.2N titanous chloride solution (MIL-STD-286, Method 601.1). Perform a blank titration on the reagents. Calculate the percent NC as follows:

$$\% \text{ NC} = \frac{46.69N\ (V-B)\ \ x\ \ 100}{E\ \ W}$$

Where:

V = Volume titanous chloride use for sample, ml
B = Volume titanous chloride used for blank, ml
W = Weight of dried sample, gm
N = Normality of titanous chloride
E = Percent nitrogen in the nitrocellulose

4.5.1.3.4 Acrylic

REAGENTS

SULFURIC ACID - 93 to 98% H_2SO_4, N2 Free

MERCURIC OXIDE (HgO) - Reagent Grade, N_2 Free

POTASSIUM SULFATE OR ANHYDROUS SODIUM SULFATE - Reagent Grade, N_2 Free

THIOSULFATE SOLUTION - 160 g $NA_2S_203.5H_20$ dissolved in one liter of H_20

SODIUM HYDROXIDE - 400 g solid NaOH dissolved in H_20 and diluted to one (1) liter

METHYL RED INDICATOR - 1 g methyl red dissolved in 200 ml ethyl alcohol

METHYLENE BLUE INDICATOR - 1 g methylene blue dissolved in 200 ml ethyl alcohol

HYDROCHLORIC OR SULFURIC ACID STANDARD SOLUTION - 0.1N

BORIC ACID SOLUTION - 4 g H_3BO_4 dissolved in 100 ml H_2O

HENGAR GRANULES

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APPARATUS

DIGESTION - Use Kjeldahl flasks of hard, moderately thick, well-annealed glass with total capacity of 500 to 800 ml. Conduct digestion over a safety-approved heating device adjusted to bring 250 ml H_2O at 25°C to a rolling boil in 5 minutes.

DISTILLATION - Use a Kjeldahl flask of 500 to 800 ml capacity fitted with a rubber stopper. The low end of an efficient scrubber bulb or trap is passed through the rubber stopper to prevent mechanical carryover of NaOH during distillation. Use rubber tubing to connect the upper end of the bulb to the condenser tube. Trap the condenser outlet to insure complete absorption of distilled NH_3 into boric acid in the receiver.

ANALYSIS

Quantitatively transfer the residue from 4.5.1.3.3 to a Kjeldahl flask. Add 3 to 4 Hengar granules, 0.7g of HgO, 15.0g of K_2SO_4 or NA_2SO_4 and 25 ml H_2SO_4 to the flask. Place the flask in an inclined position, and heat gently until the solution clears. Continue boiling for 1/2 hour minimum. Allow the solution to cool to room temperature. Add 200 ml H_2O. Cool the solution to below 25°C. Add 25 ml thiosulfate solution and mix to precipitate the mercury. Tilt the flask and add a layer of 100 ml NaOH solution without agitation. Immediately connect the flask to the distilling bulb on the condenser. With the tip of the condenser immersed in the boric acid in the receiver, rotate the flask to mix the contents thoroughly. Heat the flask until all of the NH_3 has distilled (approximately 250 ml). Titrate the boric acid solution with the standard acid using a mixed indicator containing 4 drops of methyl red, and 2 drops of methylene blue solution. Correct for the reagent blank. Calculate the percent acrylic as follows:

$$\% \text{ Acrylic} = \frac{6.05\ (V-B)\ N \times 100}{W}$$

Where:

V = Volume of standard acid for sample, ml
B = Volume of standard acid for blank, ml
N = Normality of standard acid
W = Dried sample weight, g

4.5.1.3.5 Kraft, Resin and Additives.-Calculate the percent kraft, resin and additives as follows:

$$\% \text{ Kraft} + \% \text{ Resin} + \text{Additives} = 100 - (\%NC + \%\text{Acrylic} + \%DPA)$$

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4.5.1.3.6 Report the total methylene chloride extractables $\frac{(W - C) \times 100}{W}$ and the total acetone extractables $\frac{(C - D) \times 100}{W}$.

4.5.2 Weight of Tube.-The sample tubes shall be dried in an oven (See 6.6) maintained at 160 degrees Fahrenheit ($^{\circ}F$) $\pm$ 10$^{\circ}F$ for a period of 2 hours. The tubes shall be removed from the oven and immediately place each tube in a tared container (the container may be a plastic (polyethylene) bag with a metal or plastic tie) and allowed to cool for 1 hour in the same room as the balance. Weigh to the nearest 0.01 gram. If the sample fails due to high weights, repeat the above procedure except that the oven time shall be 0.5 hours. Samples may be dried as many times as necessary or until a constant weight is obtained. The weight shall be determined as follows:

Weight of tube = B - A

A - Weight of container
B - Weight of container plus tube

5. PREPARATION FOR DELIVERY

5.1 Packing, Level B-1.-All parts shall be packed in accordance with dwg. 9282963.

5.1.1 Packing, Level B.-Not applicable.

5.2 Marking.-Marking shall be in accordance with dwg. 8799112.

6. NOTES

6.1 Ordering Data.-See MIL-A-48078.

6.2 Submission of Designs for Approval.-See 6.2.3. of MIL-A-48078. Submit equipment designs, as required to Commander, U.S. Army Armament Research and Development Command, ATTN: DRDAR-QAR, Dover, New Jersey 07801.

6.3 Packing.-The requirements herein for Level B are equivalent to requirements for full Level B use in accordance with AR700-15, 20 June 1975. The requirements under Level B-1 are intended to provide for limited storage life necessary to meet ongoing loading assembly and packing operations and to provide uniform packing from all producers of an item to all using facilities.

6.4 Submission of Ammunition Data Cards.-In addition to the normal distribution of records, when the tubes are procured by

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the Army, one copy of all ammunition data cards shall be forwarded to each of the following:

Commander
US Army Armament Research and Development Command
ATTN: DRDAR-QAR-Q
DRDAR-LCU-EP
Dover, New Jersey 07801

6.5 Drawings.-Drawings listed in Section 2 of this specification under the heading US Army Armament Research and Development Command ARRADCOM may also include drawings prepared by and identified as, Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under the cognizance of ARRADCOM.

6.6 Ovens.-Ovens utilized for drying tubes to determine weight will comply with PAD 1006.

Custodian:
Army - AR

Preparing activity:
Army - AR

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